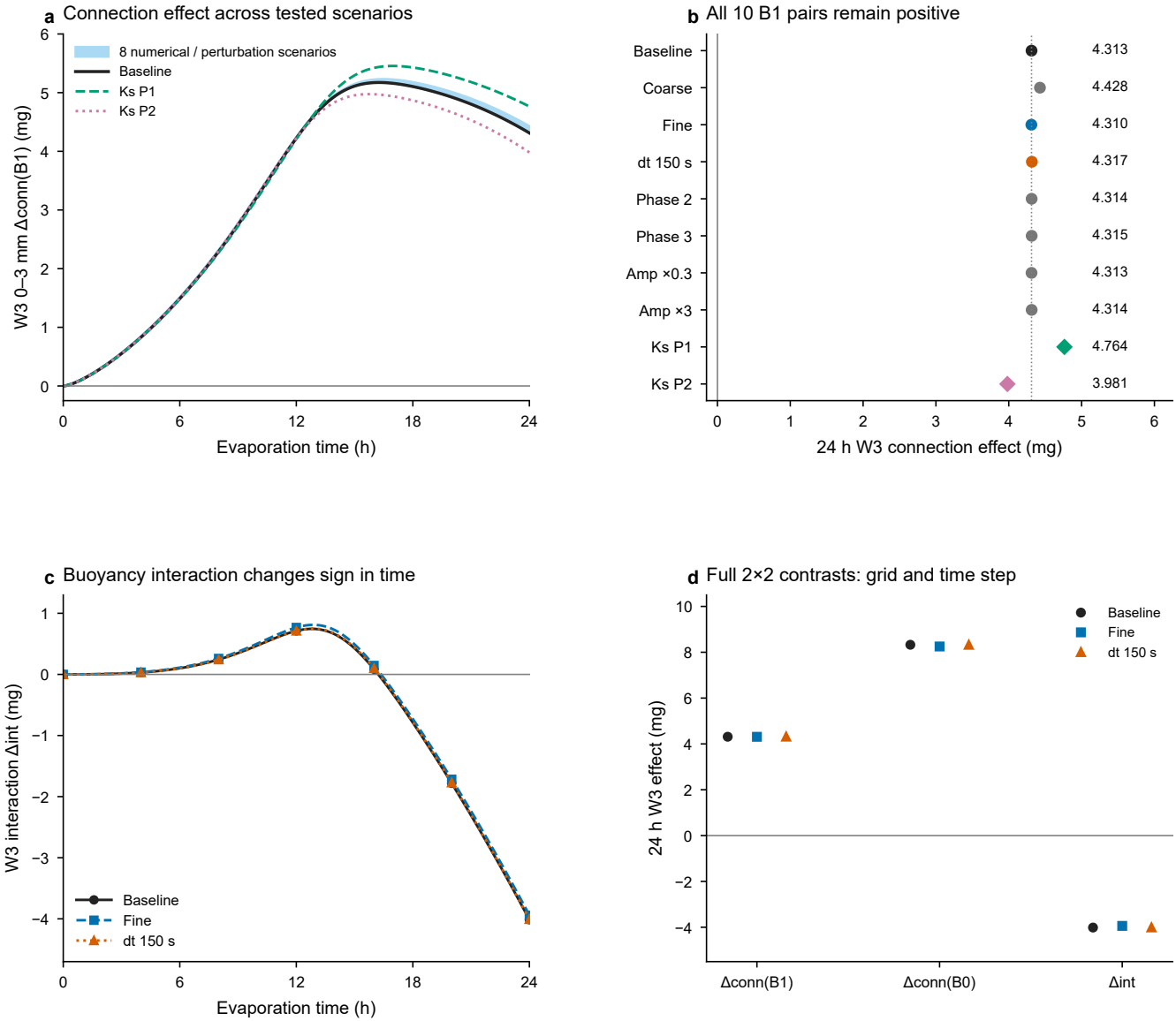
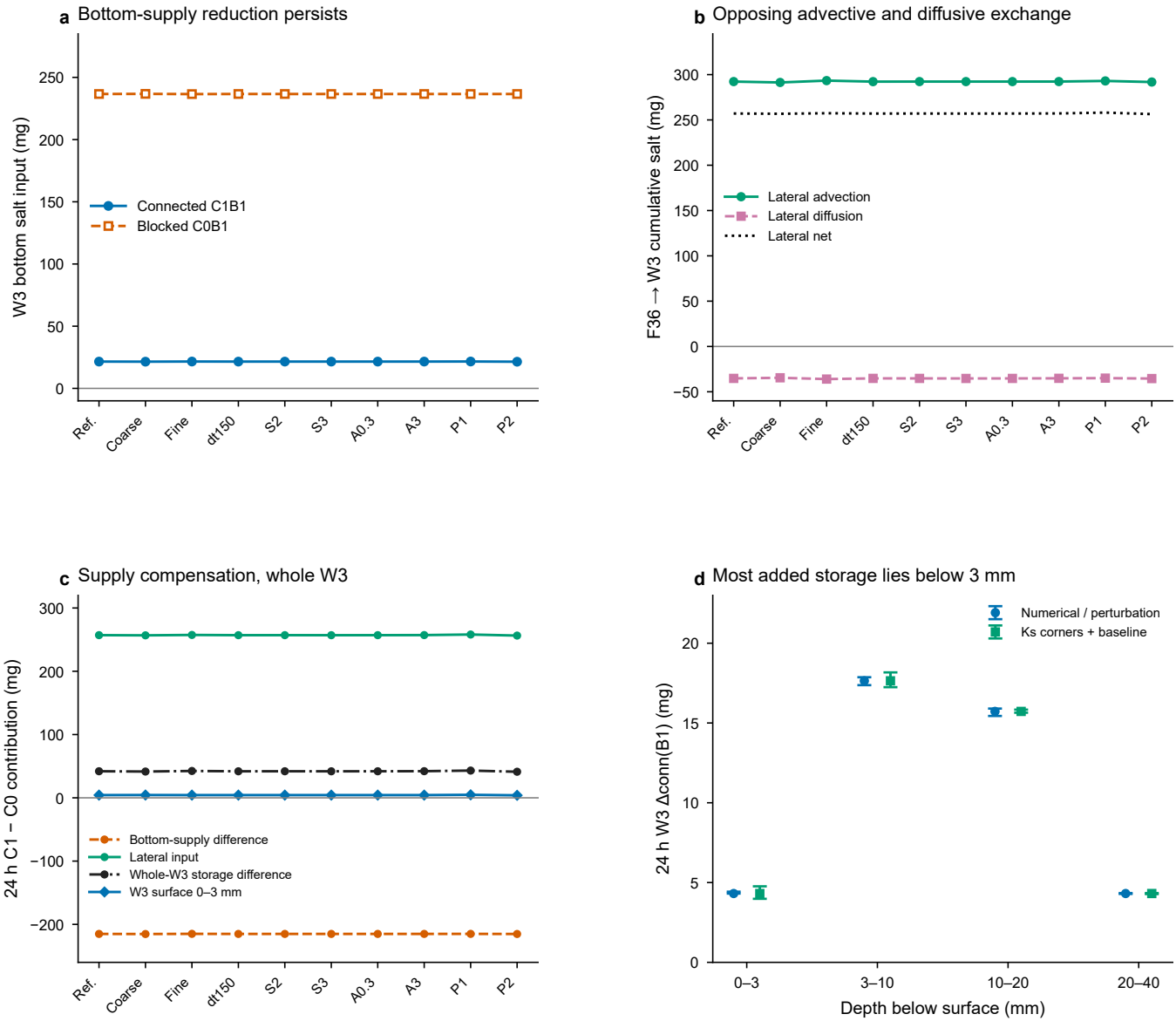
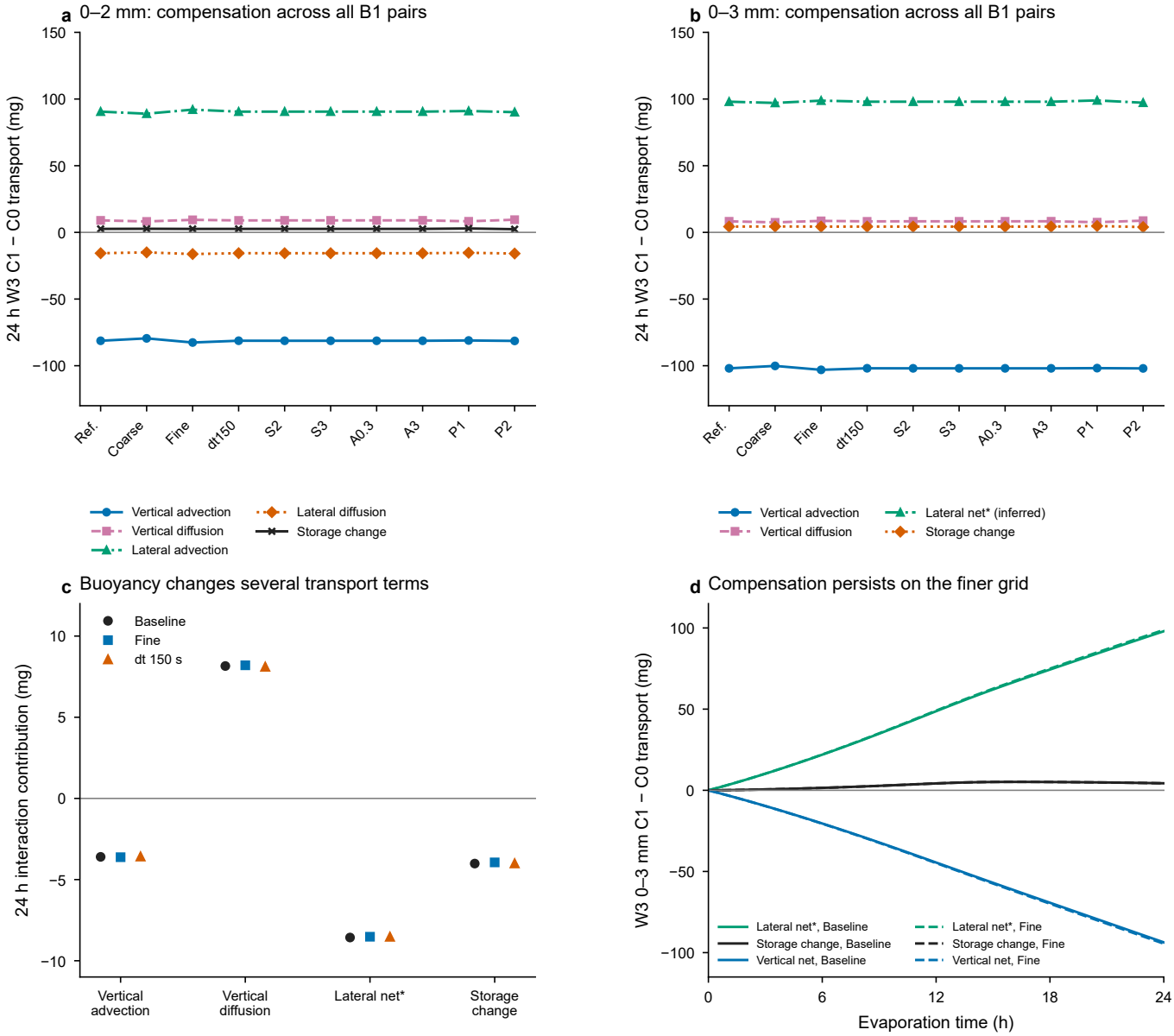


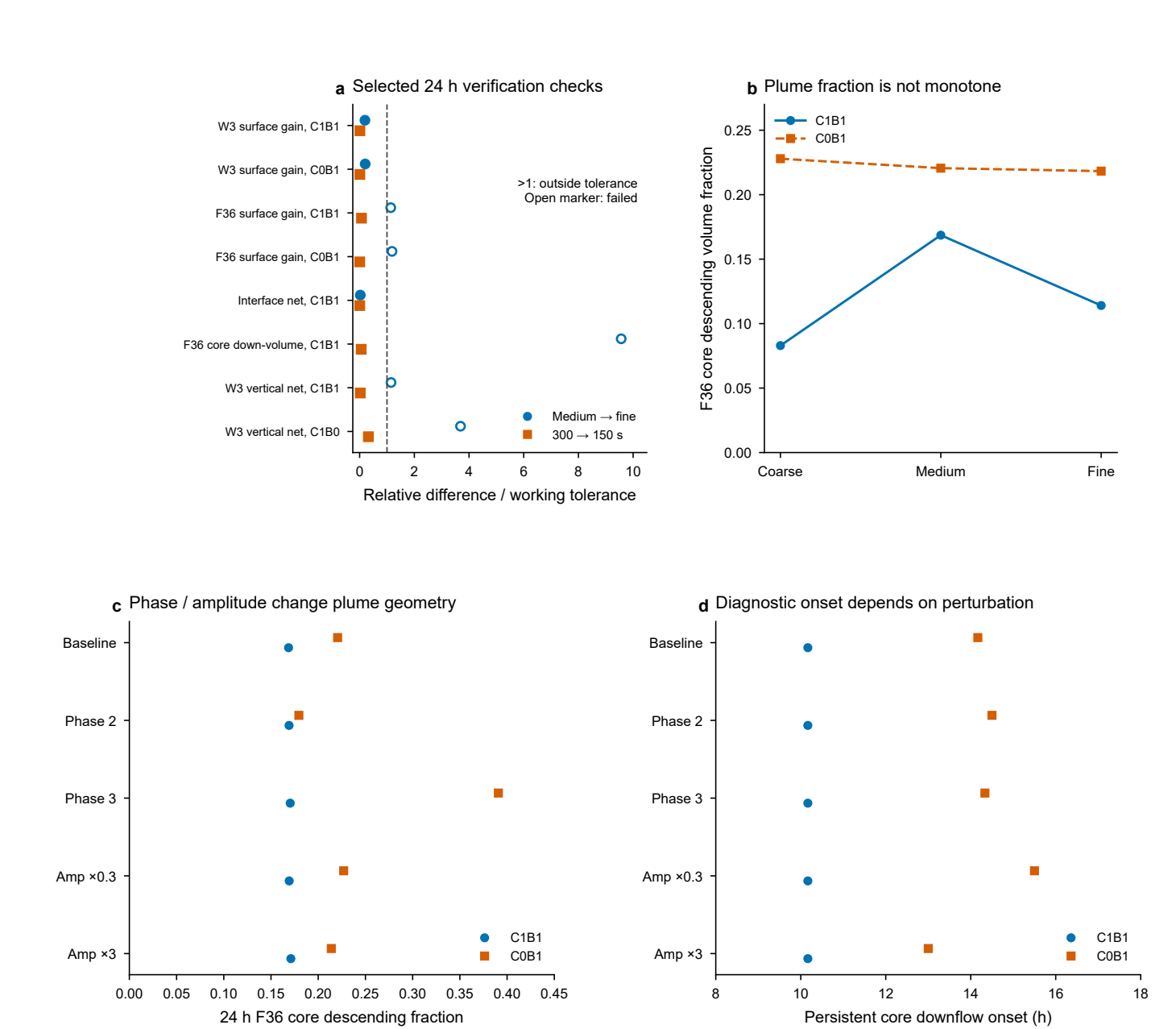
C1/C0: connected/blocked; B1/B0: salinity buoyancy retained/suppressed. Δconn(B) = C1B − C0B; Δint = Δconn(B1) − Δconn(B0).
Ranges are finite scenario ranges, not confidence intervals or a bound on all parameter uncertainty. P1/P2 are measured-Ks corners.
RH = 0; 24 h; 26 accepted cases. No new experimental validation. Numerical and perturbation envelope includes the coarse grid.



B1 only; all panels use the same 10 scenarios. S2/S3: phases 2/3; A0.3/A3: amplitude ×0.3/×3; P1/P2: conductivity corners.
Lines across scenario categories guide comparison; categories are not a continuous parameter sweep. Depth whiskers show min–max ranges, not CIs.
Whole-side budgets do not substitute for a surface-layer budget. Surface budgets are shown separately in Fig. S local transport.



Upper 2 mm: independent lateral + vertical conservative flux balance. Upper 3 mm: *lateral NET is inferred from storage minus vertical input.
Vertical terms use face weights consistent with fractional-cell storage, rather than the nearest mesh plane. No lateral 0–3 mm mode split is fabricated.
Small storage differences can coexist with large opposing transport changes. This discrete-budget result does not establish all fluxes are grid independent.
S2/S3: phases 2/3; A0.3/A3: amplitude ×0.3/×3; P1/P2: conductivity corners. NaCl-equivalent mass at 58.443 g/mol; true 24 h endpoints.



A stable W3 integral does not imply converged F36 fields. All registered checks, including unplotted rows, are supplied in SOURCE_DATA.
Surface-gain tolerance: 2%; plotted flux/volume tolerance: 5%. Vertical diagnostics retain the original registered depth interpolation.
Core descent: Darcy speed >1e-8 m/s; onset requires three consecutive 10-min frames. It is not a linear-instability or MRI-detection onset.
Filled/open markers in a indicate within/outside the working tolerance; phase/amplitude sensitivity is not solver failure.